

Clinic Cloud Migration Portfolio Closeout Report

Formal project record for Waves 0 through 6

Closure date 28 September 2026

Portfolio environment AWS Asia Pacific Hong Kong region

Data classification Sanitized portfolio record with synthetic data only

Recommended repository destination docs/portfolio-closeout/

This report closes the portfolio through Wave 6. It records the implemented platform and network work, the partially deployed Appointment API engineering, two validated migration designs that were intentionally not deployed, and the Lab and Radiology workstream terminated because its external ISP connectivity prerequisite was unavailable.

The closure is a project-management disposition, not a claim that a production clinic migration or clinical go-live occurred.

Document Control

Field	Value
Purpose	Authoritative portfolio closeout record for Waves 0 through 6
Audience	Project owner, technical reviewers, hiring reviewers, and future maintainers
Repository	clinic-cloud-iac
Recommended location	docs/portfolio-closeout/
Evidence basis	Current Git branch, tracked evidence, repository documentation, and project decision records
Excluded material	State, plans, tfvars, credentials, secret values, build archives, caches, raw payloads, and PHI or PII
Overall status	Portfolio closed through Wave 6 with differentiated terminal outcomes

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Executive Summary

The clinic cloud security engineering portfolio has reached a documented terminal state for every workstream in scope from Wave 0 through Wave 6. Waves 0, 1, and 2 closed with passing implementation and validation evidence. Wave 3 closed after engineering and testing were completed, while final deployment was intentionally deferred at the owner-controlled IAM boundary. Waves 4 and 5 validated migration architecture and security controls without deployment. Wave 6 completed discovery and then terminated because the ISP would not provide the public connectivity required for the on-premises VPN endpoint.

Wave	Final Status	Disposition
W0	CLOSED - PASS	Implemented foundation
W1	CLOSED - PASS	Implemented and validated
W2	CLOSED - PASS	Implemented and validated
W3	ENGINEERING CLOSED - DEPLOYMENT DEFERRED	Partially deployed; engineering validated
W4	DESIGN VALIDATED - NOT DEPLOYED	Design only
W5	DESIGN VALIDATED - NOT DEPLOYED	Design only
W6	TERMINATED - EXTERNAL CONNECTIVITY PREREQUISITE UNAVAILABLE	Discovery completed; implementation terminated

The portfolio demonstrates a controlled progression from account and platform foundation to Terraform lifecycle management, existing-network adoption, serverless security engineering, workload migration design, and hybrid-connectivity due diligence. It also records decisions not to deploy when controls, ownership, evidence, cost, or external prerequisites were insufficient.

Closure conclusion

Waves 0 through 6 are formally closed for portfolio purposes. The accurate public description is that the portfolio completed implemented, deployment-deferred, design-validated, and externally terminated workstreams. It must not be described as a completed production clinic migration, a production-ready clinical platform, or an operational hybrid deployment.

Project Scope and Evidence Boundary

Project purpose

The project was created as a synthetic healthcare cloud migration and security engineering portfolio. It used a nonproduction AWS account and Terraform to demonstrate how a clinic environment could be assessed, governed, adopted into infrastructure as code, secured, tested, and closed. The work emphasized private networking, encryption, controlled identity, recoverability, auditability, cost awareness, and honest acceptance criteria.

Evidence boundary

- The current repository contains formal tracked evidence for Wave 2 through Wave 5, detailed documentation and tests for Wave 3 through Wave 5, and Git history for the major engineering checkpoints.
- Wave 0 and the FSx recovery incident are supported by project decision records and later infrastructure discovery. The current branch does not contain a dedicated Wave 0 closeout artifact.
- Wave 6 discovery facts are supported by the final read-only discovery record in the project conversation. No Wave 6 repository or AWS mutation occurred.
- Validation statements distinguish static tests, Terraform validation, read-only discovery, partial deployment, and live operation. A design pass is never treated as a deployment pass.

Data handling

The project uses synthetic or mock data only. The closeout package omits secret values, credentials, private keys, state files, binary Terraform plans, tfvars, raw queue payloads, build ZIPs, PHI, PII, and local credential caches. Account identifiers, resource identifiers, local user paths, and public IP addresses are redacted in the packaged supporting copies.

Architecture and Security Evolution

Foundation to governed nonproduction

Wave 0 established the organizational and account direction: a separated nonproduction environment, workforce access through IAM Identity Center, Hong Kong region validation, service availability discovery, and a private clinic platform baseline. The early environment included private network services such as Directory Service and FSx, with recovery and monitoring work treated as explicit engineering gates rather than assumed controls.

Terraform lifecycle and protected state

Wave 1 converted the project into a repeatable infrastructure-as-code workflow. It covered format, validate, saved-plan review, controlled apply, independent verification, drift detection, import and state ownership, remote S3 state, versioning, encryption, public-access blocking, TLS-only policy, and native state locking. Temporary plans and local variable files were classified as local artifacts and excluded from source control.

Network adoption and private service access

Wave 2 adopted and reconciled existing network resources into a reusable network module. The final module represented 18 network resources and produced a zero-change plan. The design kept private subnets without an internet default route, scoped the S3 gateway endpoint to private route tables, retained the Directory Service security group as an external dependency, and deferred broader endpoint creation until workload requirements justified it.

Serverless appointment boundary

Wave 3 introduced a public appointment intake boundary while preventing direct access to clinical databases, FSx, and Directory Service. The design used AWS WAF, API Gateway, Lambda, DynamoDB, SQS and a dead-letter queue, EventBridge reconciliation, CloudWatch logging and alarms, SNS routing, HMAC authentication, timestamp and nonce checks, replay prevention, atomic idempotency, workflow state controls, retries, exception handling, and manual-review states. Logs were designed to exclude request bodies, patient values, signatures, tokens, and secret material.

Workload migration designs

Wave 4 modeled a private Pharmacy application on Windows EC2 with private RDS for SQL Server, TLS client requirements, narrow security-group flows, Session Manager administration, external identity and secret ownership, provider integration alternatives, backups, restore validation, and synthetic prescription reconciliation. Wave 5 extended these patterns to a shared Clinic and EMR MySQL workload using private EC2, private RDS MySQL, and a temporary private DMS full-load plus CDC path. Wave 5 added clinical RBAC, break-glass controls, audit continuity, data integrity checks, cutover and rollback sequencing, and conditional FSx reference consistency.

Hybrid integration discovery

Wave 6 planned the Lab and Radiology hybrid path around encrypted protocols, strict routes, patient matching, checksums, acknowledgements, bounded retries, idempotency, downtime queueing, manual fallback, reconciliation, and staged alerting. Read-only discovery found an existing VGW and dynamic BGP VPN design, but both tunnels were down, accepted routes were zero, and private route tables had neither

VGW propagation nor explicit on-premises routes. The ISP would not provide the required public connectivity, so implementation was terminated.

Wave by Wave Record

W0 Project and account foundation

Final status CLOSED - PASS

Delivery classification Implemented foundation

Landing-zone direction, nonproduction account, identity access, regional/service discovery, private platform baseline, and recovery groundwork.

- Established the nonproduction account and region access path.
- Validated workforce identity and read-only service discovery patterns.
- Built the initial private platform direction around network isolation, Directory Service, FSx, encryption, backup, logging, and later migration gates.
- Recorded the accidental deletion of an empty FSx file system as an incident and recreated the service without claiming data restoration.

W1 Terraform and IaC foundation

Final status CLOSED - PASS

Delivery classification Implemented and validated

Terraform workflow, state protection, import/lifecycle, drift reconciliation, remote S3 state, locking, and repository hygiene.

- Created the protected S3 backend with versioning, encryption, block-public-access, TLS-only policy, force-destroy disabled, and lifecycle protection.
- Migrated the nonproduction root from local state to remote state and validated locking through competing access.
- Demonstrated import, state ownership removal, first managed resource creation, controlled drift, and drift reconciliation.
- Closed with recursive formatting, validation, zero-drift plans, and repository hygiene checks.

W2 Network foundation and reconciliation

Final status CLOSED - PASS

Delivery classification Implemented and validated

Existing network adoption, modular refactor, private routing, security-group reconciliation, and an S3 gateway endpoint scoped to private route tables.

- Reconciled existing VPC, subnet, route-table, internet gateway, route association, security-group, and endpoint ownership.
- Refactored the adopted network into a module with 18 resources and no residual root-level network resources.
- Created and verified one S3 gateway endpoint associated only with the two private route tables.
- Closed with Terraform validation, plan exit code zero, no changes, and no tracked or untracked risk artifacts in the repository.

W3 Appointment API workflow

Final status ENGINEERING CLOSED - DEPLOYMENT DEFERRED

Delivery classification Partially deployed; engineering validated

Secure serverless workflow engineered with 205 passing tests. Thirty Terraform-managed resources exist; the reviewed remaining 21 non-IAM resources were not deployed.

- Completed architecture, HMAC and replay controls, edge hardening, workflow reconciliation, monitoring, and adversarial validation.
- Passed 205 application, contract, unit, security, privacy, and Terraform-invariant tests.
- Retained 30 deployed Terraform-managed Wave 3 resources; did not deploy the final reviewed 21-resource non-IAM remainder.
- Moved HMAC secret and four IAM roles to external owner-controlled governance after provider and permission incidents.
- Deleted the saved plan at closeout and marked it unauthorized for reuse.

W4 Pharmacy migration

Final status DESIGN VALIDATED - NOT DEPLOYED

Delivery classification Design only

Private Windows EC2 and RDS SQL Server design, provider integration alternatives, recovery design, and 19 passing static design tests. No AWS mutation or migration.

- Passed Terraform formatting and validation and 19 static design tests.
- Modeled no public EC2 or RDS address, no routine RDP, and SQL access only from the application security group.
- Documented HTTPS, callback, and SFTP provider patterns; prohibited FTP.
- Performed no AWS calls, Terraform plan, apply, live cutover, or migration.

W5 Clinic and EMR shared MySQL migration

Final status DESIGN VALIDATED - NOT DEPLOYED

Delivery classification Design only

Private EC2, RDS MySQL, temporary private DMS full-load plus CDC design, RBAC/UAT and integrity design, and 34 passing static design tests. No AWS mutation or migration.

- Passed Terraform formatting and validation and 34 static design tests.
- Modeled private RDS MySQL and DMS full-load plus CDC with certificate verification, validation, bounded logging, and external secrets.
- Added clinical RBAC, break-glass governance, integrity and CDC tests, audit continuity, FSx reference checks, cutover, rollback, and isolated restore design.
- Performed no AWS mutation, DMS task, database migration, clinical UAT, cutover, or rollback.

W6 Lab and Radiology hybrid integration

Final status TERMINATED - EXTERNAL CONNECTIVITY PREREQUISITE UNAVAILABLE

Delivery classification Discovery completed; implementation terminated

Read-only discovery found both existing VPN tunnels down, zero accepted routes, and no private-route-table path to on-premises networks. ISP-side public connectivity was unavailable.

- Identity gate passed for the correct nonproduction account.
- Existing VPN was available as an AWS object but both tunnels were down and accepted zero routes.
- No Transit Gateway existed; the private route tables exposed no usable on-premises path.
- Security-group and logging gaps were identified, but no remediation was attempted because the external connectivity prerequisite failed.
- No Terraform, AWS, repository, migration, or clinical-validation mutation occurred.

Incident and Recovery Lessons

Accidental FSx deletion

An empty nonproduction FSx file system was accidentally deleted. Because the file system contained no material workload data, the project recreated it rather than claiming a backup restore. The incident exposed a key recovery distinction: FSx automatic backups and centrally governed AWS Backup coverage serve different purposes, and an explicit backup resource assignment does not automatically follow a replacement file-system identifier.

Incident stage	Record
Cause	Operator or configuration error caused deletion of an empty nonproduction FSx file system.
Impact	No production or clinical data loss; service identity changed after recreation.
Recovery	Recreate the empty file system, reconnect Directory and network dependencies, and verify backup assignment for the new resource.
Validation lesson	A recreation is not a restore. Recovery evidence should verify SMB access, permissions, checksums, backup capture, and recovery timing when data exists.
Prevention	Use lifecycle protection, deletion protection where supported, stable DNS abstraction, explicit approvals, protected backup selection, and recurring assignment checks.

Wave 3 partial apply and provider recovery

Wave 3 experienced partial Terraform apply attempts and a provider stall around Secrets Manager create or refresh behavior. The response preserved the state boundary, stopped repeated mutation attempts, treated saved plans as single-use artifacts, and refactored the HMAC secret from Terraform ownership to an external ARN contract. Later IAM create-role failures led to a second ownership refactor: four execution and logging roles became external prerequisites.

- Do not assume a failed apply is atomic; inspect persisted resources and state before any recovery plan.
- Bind approval to an exact plan hash and change shape, then invalidate and delete the plan when ownership or configuration changes.
- Stop instead of widening permissions when a separation-of-duties boundary is deliberate.
- A resource ARN contract can preserve application design while keeping secret values and IAM administration outside Terraform.
- Close the wave honestly when engineering is complete but operational prerequisites remain unavailable.

Expired SSO during Wave 6 discovery

The first Wave 6 discovery attempt stopped at the mandatory identity gate when the SSO token expired and refresh failed. No further AWS commands were run. After the owner reauthenticated, the read-only identity

gate passed and discovery resumed. The lesson is that account identity is a prerequisite, not a routine preamble that can be bypassed.

Validation and Testing Evidence

Area	Evidence	Result	Boundary
Wave 1	Terraform format, validate, plans, import, drift, state migration, locking, repository checks	PASS	Nonproduction training and platform foundation
Wave 2	Terraform validate; zero-change plan; 18-resource network module; risk-artifact review	PASS	Implemented nonproduction network ownership
Wave 3	205 tests; Terraform format and validate; negative/security tests; package hash; sensitive-file review	PASS	Engineering pass; live deployment incomplete
Wave 4	19 static design tests; format and validation; sensitive design review	PASS	Design only; no AWS or migration test
Wave 5	34 static design tests; format and validation; sensitive-file and credential-signature scan	PASS	Design only; no DMS task, UAT, or cutover
Wave 6	Read-only identity, VPN, route, security-group, NACL, logging, and cost discovery	PREREQUISITE FAILED	No end-to-end hybrid validation

Wave 3 security coverage

- HMAC signature, timestamp window, nonce, replay, key selection, and fail-closed behavior.
- Atomic idempotency, workflow state transitions, queue failure handling, worker leases, retries, reconciliation, and manual-review outcomes.
- Schema limits, request-size controls, WAF and method boundaries, privacy-preserving errors, and log minimization.
- Terraform invariants for private clinical boundaries, external IAM and secret ownership, encryption, logging, alarms, and least-privilege transport policies.

Evidence discipline

The evidence distinguishes implementation from design. A successful Terraform validation proves configuration coherence, not cloud deployment. A static test proves a modeled invariant, not vendor compatibility or clinical acceptance. A read-only discovery finding proves observed state at the collection time, not durable operation. These distinctions are retained in every final status.

IAM and Secrets Governance

The owner retained IAM administration and declined to delegate or bootstrap broad role-creation authority. That decision set the final Wave 3 deployment boundary and became the default ownership model for the Wave 4 and Wave 5 designs.

Control area	Decision	Effect
Human access	Temporary SSO credentials and account identity checks	Avoid long-lived credentials in Terraform and project files
IAM administration	Owner controlled	No AdministratorAccess workaround and no automatic role creation
PassRole	Exact role and service restriction required	Future deployment must authorize only reviewed roles and target services
HMAC secret	External container and value lifecycle	Terraform consumes only an ARN; no secret value enters code, state, plan, logs, or evidence
KMS and backup roles	External governance in later designs	Key administration and backup deletion remain separated from workload deployment
Break glass	Named, time-limited, alerted, reasoned, reviewed	No standing or shared emergency database account

Wave 3 remaining gates

- Four external IAM roles would need to exist before deployment resumes.
- A deployment identity would need exact, service-restricted iam:PassRole.
- Synthetic HMAC material remains unprovisioned.
- A human alert destination remains unconfigured.
- Live API, queue, monitoring, privacy, and recovery controls remain unvalidated.

Cost Control Decisions

Cost gates influenced technical scope. The portfolio favored the smallest design that could demonstrate the intended control and avoided creating resources whose requirements were unproven.

Decision	Cost effect	Engineering rationale
S3 gateway endpoint instead of broad endpoint catalogue	No interface-endpoint hourly charges	Created only the justified private S3 path
No speculative NAT Gateway	Avoided fixed hourly and data-processing cost	Private workloads and provider paths were not yet evidenced
Wave 4 Single-AZ conceptual baseline	Reduced Windows and SQL Server pilot cost	Design validation only; not a production availability recommendation
Wave 5 temporary DMS design	Migration runtime would exist only during a governed migration	No DMS resource was created
External secrets and IAM	Avoided unsafe workarounds; not primarily a cost decision	Preserved owner governance and limited deployment scope
Existing VPN reuse considered	Avoided a second VPN, but existing connection charges may continue	A separate decommission decision is required if the VPN has no future purpose

Deferred cost decisions

- SQL Server edition, Windows sizing, licensing, storage, and Multi-AZ requirements need vendor and workload evidence.
- RDS MySQL, EC2, temporary DMS, endpoints, hybrid transfer, and optional FSx usage need measured sizing.
- The 30 deployed Wave 3 resources should be retained or decommissioned only after a separate inventory, cost, evidence-retention, and approval decision.
- The unusable VPN may continue to incur connection and public IPv4-related charges; deletion was not authorized by this closeout.

Portfolio and Resume Framing

Accurate portfolio description

Healthcare cloud security engineering migration portfolio completed through Wave 6 across implemented, deployment-deferred, design-validated, and externally terminated workstreams.

Resume bullet options

- Engineered and documented a synthetic clinic AWS migration portfolio spanning account governance, Terraform state, existing-network adoption, secure serverless workflows, workload migration design, and hybrid-connectivity discovery.
- Built a security-focused Appointment API design with WAF, API Gateway, Lambda, DynamoDB, SQS, HMAC authentication, replay protection, atomic idempotency, reconciliation, privacy controls, and 205 passing tests; closed deployment as deferred at an owner-controlled IAM boundary.
- Reconciled an existing AWS network into an 18-resource Terraform module, verified zero drift, and deployed a private-route-table S3 gateway endpoint without expanding public routing.
- Designed nondeployed Pharmacy and Clinic or EMR migration patterns using private EC2 and RDS, DMS full load plus CDC, TLS, least-privilege security groups, externally governed secrets and keys, backup and restore validation, cutover, rollback, and integrity controls.
- Assessed Site-to-Site VPN and BGP readiness for Lab and Radiology integration; terminated implementation after read-only discovery found both tunnels down, zero accepted routes, and an unavailable ISP-side public connectivity prerequisite.
- Converted an accidental empty FSx deletion and Wave 3 partial-apply failures into documented recovery, ownership, plan-integrity, and change-control lessons without overstating production recovery or deployment completion.

Interview framing

The strongest narrative is not that every proposed workload was deployed. It is that the project used evidence and governance to decide when to implement, when to stop, when to redesign ownership, and when to terminate a workstream. The candidate should be able to explain the difference between Terraform validation, design validation, partial deployment, live operational acceptance, and production readiness.

Claims to avoid

- Completed a full clinic or hospital migration to AWS.
- Deployed a production Appointment API.
- Migrated Pharmacy or Clinic and EMR databases.
- Completed a live DMS cutover or clinical UAT.
- Established a functioning Lab and Radiology VPN.
- Achieved healthcare compliance certification or production go-live approval.

Limitations and Residual Risks

Area	Limitation or risk	Closure treatment
Production use	No production workload, production data, or clinical authorization	Portfolio only
Wave 0 evidence	No dedicated Wave 0 artifact in the current Git branch	Use project records and later discovery; retain as a documentation limitation
Wave 3	Final 21 resources, IAM roles, HMAC value, alert destination, and live controls incomplete	Deployment deferred
Wave 4	Vendor, schema, provider contract, connectivity, and clinical requirements absent	Design validated only
Wave 5	Source inventory, route, RPO/RTO, CDC threshold, compatibility, and clinical UAT absent	Design validated only
Wave 6	ISP public connectivity unavailable; tunnels down; no accepted prefixes or private routes	Workstream terminated
Backups	FSx retention was observed as zero days during Wave 6 discovery	Open resilience concern; no mutation in closeout
Logging	One VPN tunnel lacked logging; retention, KMS, metric filters, and alarms were incomplete	Recorded as discovery finding
Cost	Deployed Wave 3 and existing VPN resources may continue to incur charges	Separate inventory and decommission approval required

Final Closure Statement

The clinic-cloud-iac portfolio is formally closed through Wave 6 as of 28 September 2026. W0, W1, and W2 are CLOSED - PASS. W3 is ENGINEERING CLOSED - DEPLOYMENT DEFERRED. W4 and W5 are DESIGN VALIDATED - NOT DEPLOYED. W6 is TERMINATED - EXTERNAL CONNECTIVITY PREREQUISITE UNAVAILABLE.

This closure preserves the implemented infrastructure and code history, the validated designs, the incident and recovery lessons, the owner-controlled IAM and secrets boundary, the cost decisions, and the evidence limitations. It authorizes no Terraform apply, AWS mutation, IAM change, secret provisioning, VPN change, migration, cutover, decommissioning, Git push, or production claim.

Any future resumption must begin as a new governed phase with current identity verification, refreshed discovery, new requirements, a new Terraform plan where applicable, security and cost review, explicit owner approval, and new evidence. No saved plan referenced by this closeout is authorized for reuse.

Appendices

Appendix A Final Status Register

Wave	Topic	Final Status	Delivery Class	Closeout Basis
W0	Project and account foundation	CLOSED - PASS	Implemented foundation	Landing-zone direction, nonproduction account, identity access, regional/service discovery, private platform baseline, and recovery groundwork.
W1	Terraform and IaC foundation	CLOSED - PASS	Implemented and validated	Terraform workflow, state protection, import/lifecycle, drift reconciliation, remote S3 state, locking, and repository hygiene.
W2	Network foundation and reconciliation	CLOSED - PASS	Implemented and validated	Existing network adoption, modular refactor, private routing, security-group reconciliation, and an S3 gateway endpoint scoped to private route tables.
W3	Appointment API workflow	ENGINEERING CLOSED - DEPLOYMENT DEFERRED	Partially deployed; engineering validated	Secure serverless workflow engineered with 205 passing tests. Thirty Terraform-managed resources exist; the reviewed remaining 21 non-IAM resources were not deployed.
W4	Pharmacy migration	DESIGN VALIDATED - NOT DEPLOYED	Design only	Private Windows EC2 and RDS SQL Server design, provider integration alternatives, recovery design, and 19 passing static design tests. No AWS mutation or migration.
W5	Clinic and EMR shared MySQL migration	DESIGN VALIDATED - NOT DEPLOYED	Design only	Private EC2, RDS MySQL, temporary private DMS full-load plus CDC design, RBAC/UAT and integrity design, and 34 passing static design tests. No AWS mutation or migration.
W6	Lab and Radiology hybrid integration	TERMINATED - EXTERNAL CONNECTIVITY PREREQUISITE UNAVAILABLE	Discovery completed; implementation terminated	Read-only discovery found both existing VPN tunnels down, zero accepted routes, and no private-route-table path to on-premises networks. ISP-side public connectivity was unavailable.

Appendix B Key Repository Evidence

- README.md and AGENTS.md for scope, project status, safety, and permanent nondeployment boundaries.
- environments/nonprod/evidence/W2.8/w2.8.6-wave2-closeout.json.
- environments/nonprod/evidence/W3.0 through W3.8, especially the deployment-deferred closeout.
- environments/nonprod/evidence/W4/W4_W4-A_20260928_Pharmacy-Design-Validation.json.
- environments/nonprod/evidence/W5/W5_W5-A_20260928_Clinic-EMR-Design-Validation.json.
- application/appointment-api documentation and 205-test suite.
- application/pharmacy and application/clinic-emr design, security, runbook, and test artifacts.
- Git history from the Wave 1 foundation through the Wave 5 design closeout.
- Project decision records for the FSx incident, Wave 3 recovery, owner-controlled IAM decision, and Wave 6 termination.